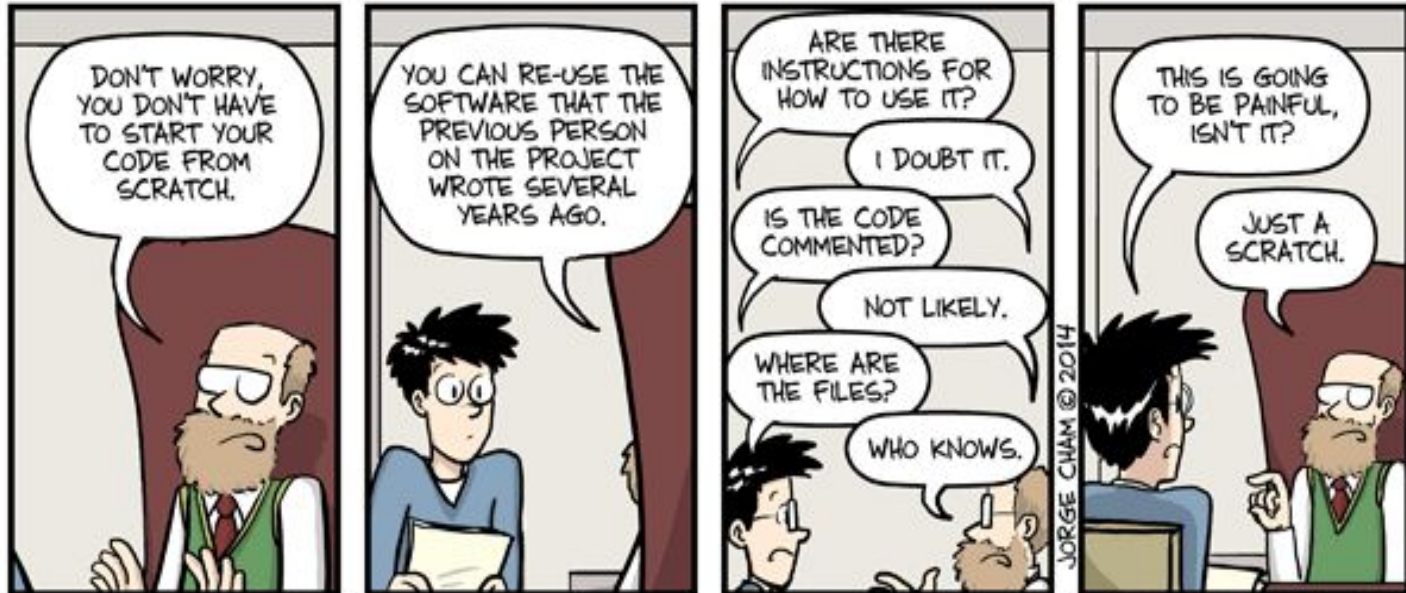


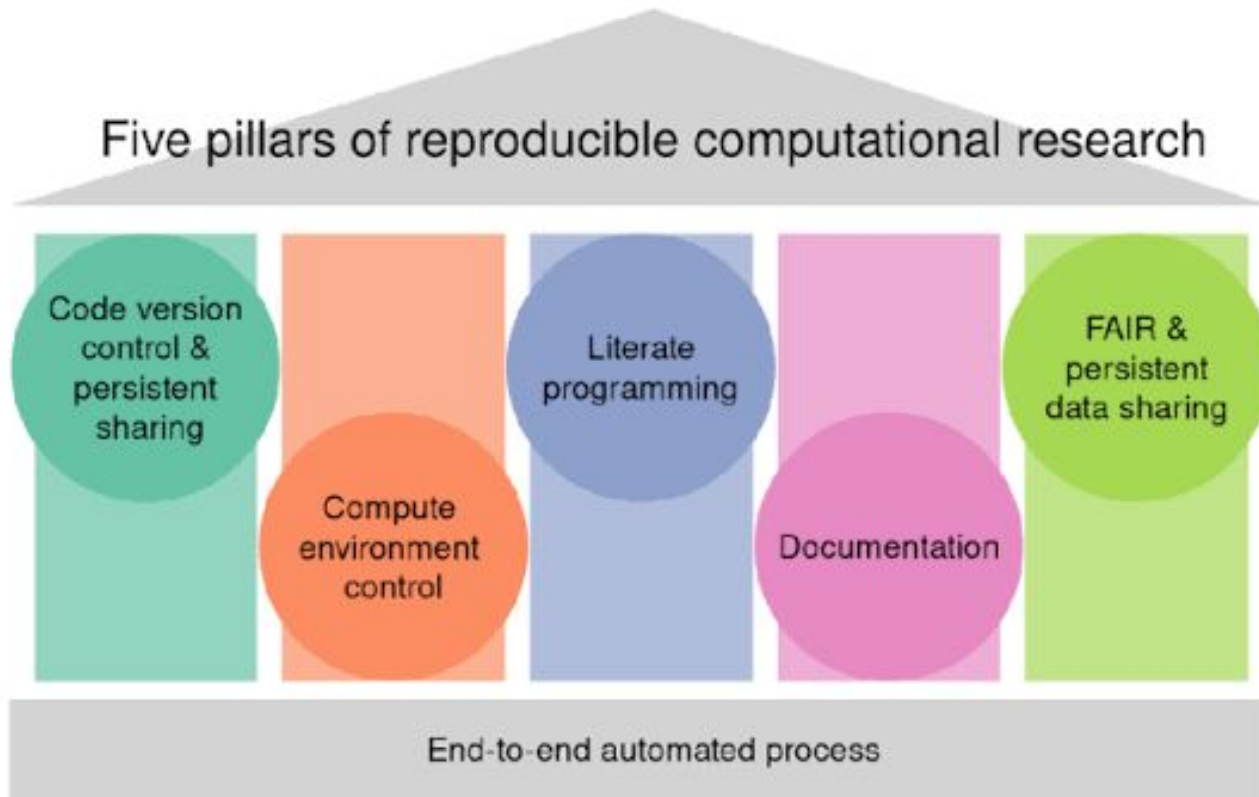
# Computational Reproducibility in Practice

Dr Cyril Pernet, Dr Melanie Ganz, Dr Michael Hanke



# Intended Learning Outcomes

- Define what computational reproducibility is and how this is achieved
- Understand what virtual environments, containers, and virtual machines are and explain how they differ
- Set a computational environment
- Containerised an application/analysis



The ability to recreate the exact results of a study using the original data, code, and analysis conditions - Ziemann , Brief Bioinformatics 2023

# It matters more than you think

Growth in a Time of Debt

Carmen M. Reinhart and Kenneth S. Rogoff

NBER Working Paper No. 15639

January 2010, Revised January 2010

JEL No. E2,E3,E6,F3,F4,N10

## ABSTRACT

We study economic growth and inflation at different levels of government and external debt. Our analysis is based on new data on forty-four countries spanning about two hundred years. The dataset incorporates over 3,700 annual observations covering a wide range of political systems, institutions, exchange rate arrangements, and historic circumstances. Our main findings are: First, the relationship between government debt and real GDP growth is weak for debt/GDP ratios below a threshold of 90 percent of GDP. Above 90 percent, median growth rates fall by one percent, and average growth falls considerably more. We find that the threshold for public debt is similar in advanced and emerging economies. Second, emerging markets face lower thresholds for external debt (public and private)—which is usually denominated in a foreign currency. When external debt reaches 60 percent of GDP, annual growth declines by about two percent; for higher levels, growth rates are roughly cut in half. Third, there is no apparent contemporaneous link between inflation and public debt levels for the advanced countries as a group (some countries, such as the United States, have experienced higher inflation when debt/GDP is high). The story is entirely different for emerging markets, where inflation rises sharply as debt increases.

- Used by politicians to explain we need austerity measures to reduce debt
- A reanalysis of the available upon request excel spreadsheet showed coding errors, selective exclusion of available data (australia, new zealand, canada), and unconventional weighting of summary statistics.

# It matters more than you think

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## ABSTRACT

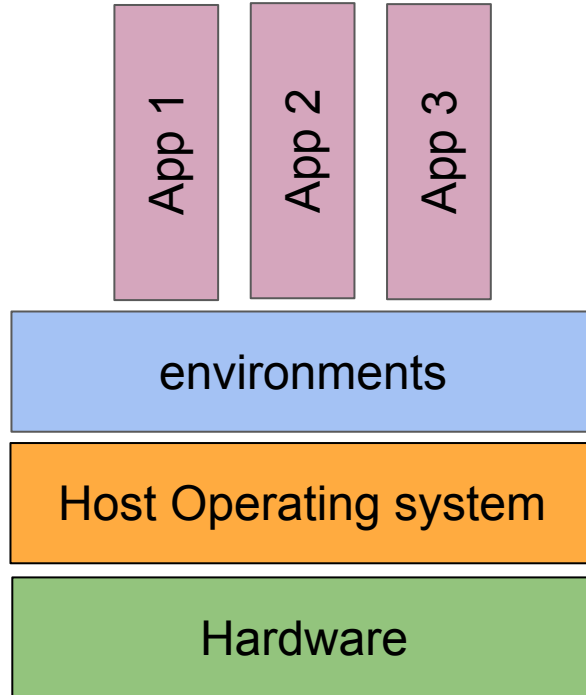
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- When properly calculated, the average real GDP growth rate for countries carrying a public-debt-to-GDP ratio of over 90 percent is actually 2.2 percent, not -0.1 percent

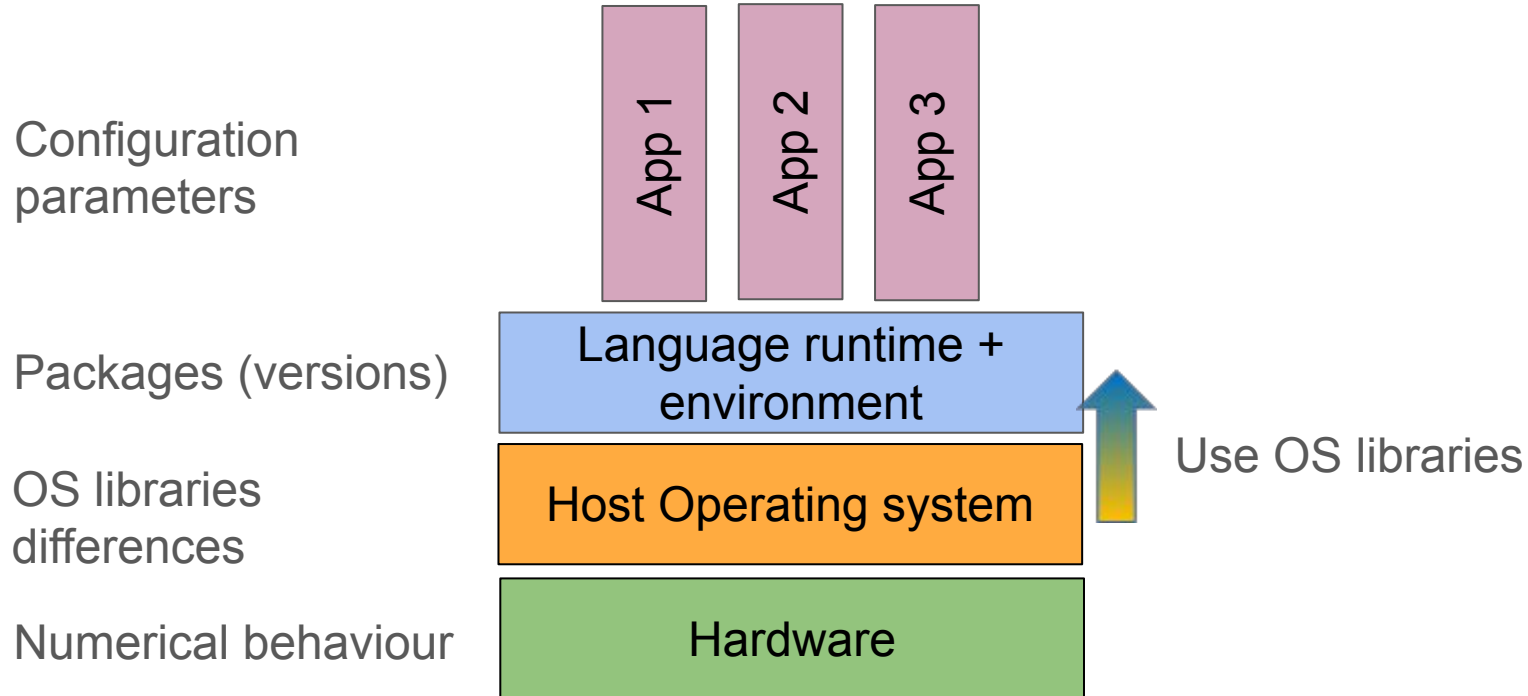
→ a proper record of hardware, software configuration and workflow is necessary to reproduce results

# Virtual environments, containers and virtual machines

What is changing  
from one person/PC  
to the next, at each  
level of this stack?



# Virtual environments, containers and virtual machines



# Teachers' notes

**Hardware level:** CPU/GPU precision differences, but also CPU versus GPU algorithms, parallel operation order, nondeterministic kernels, x86 versus ARM, or different drivers.

**Virtual environment:** Python or R is the runtime; the environment contains a particular runtime and its installed packages. A Python virtual environment is built on a base Python installation and isolates Python packages, but it does not isolate the host OS or all native system libraries

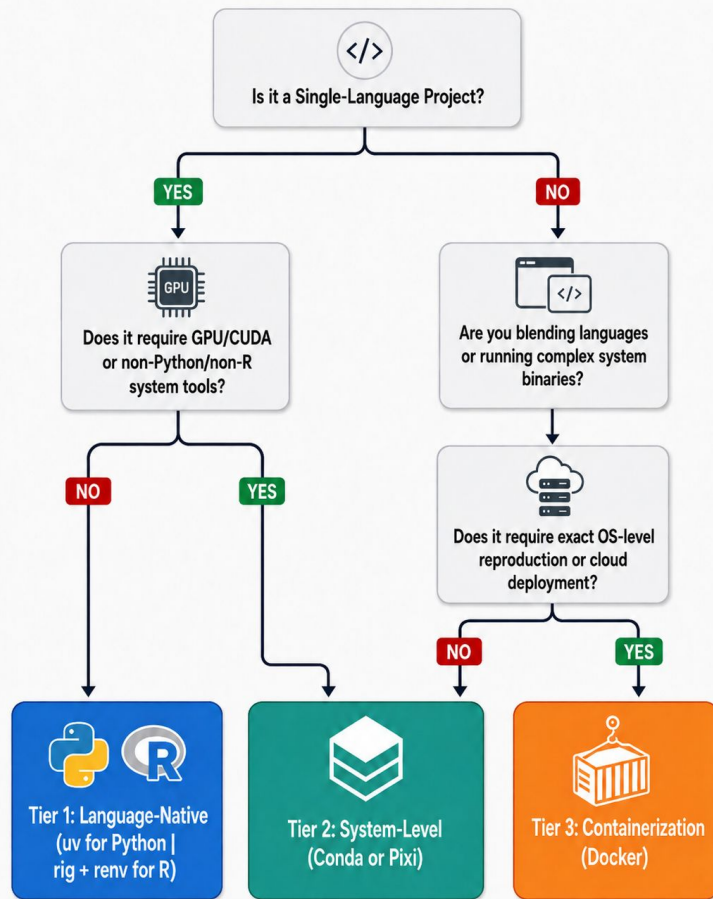
# Virtual Environment(s)

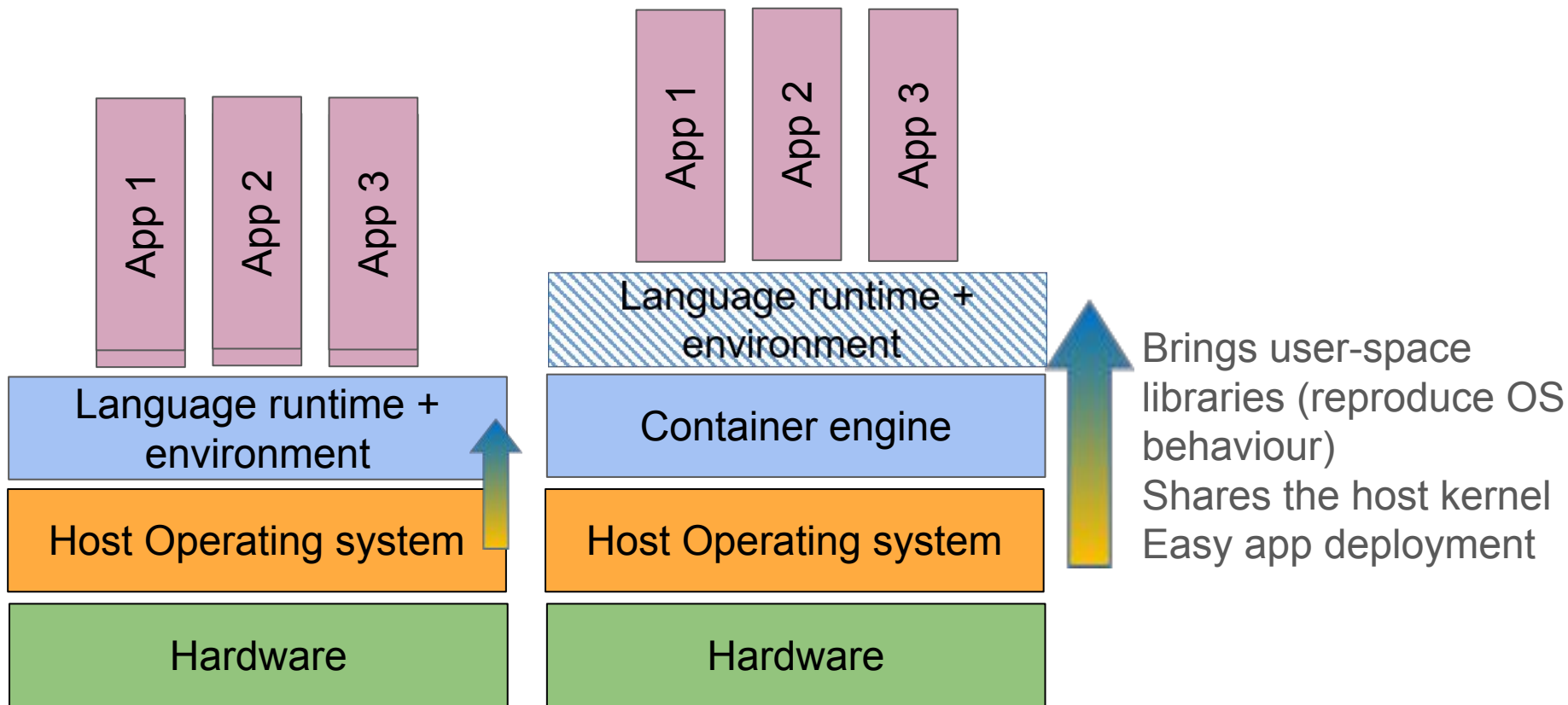
This can be done ‘automatically’ using tools such as uV (python), rig (R), or conda (python, R, etc).

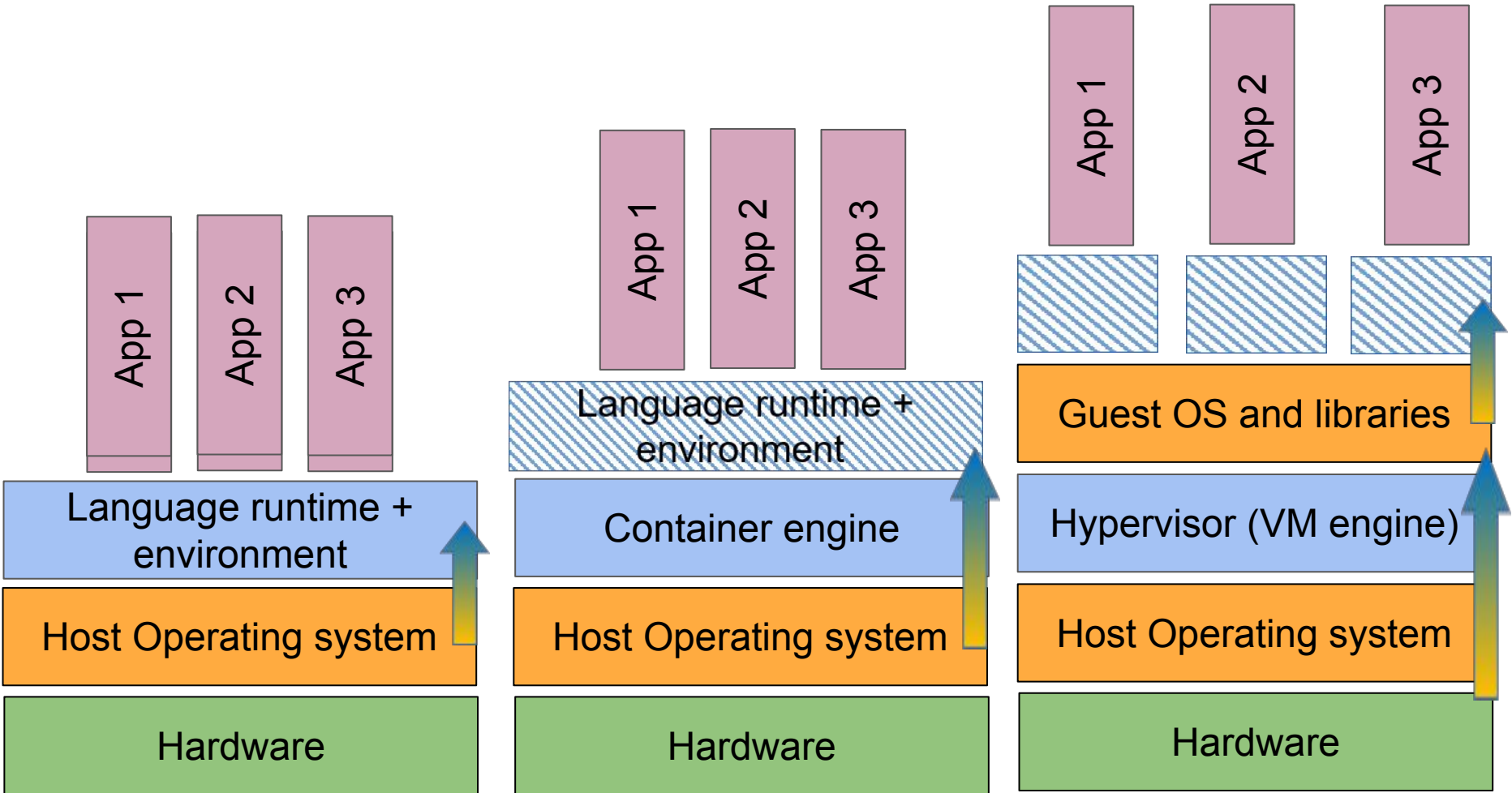
Which environment manager would you need?

## Which Environment Management Tier Should You Use?

Decision tree for Python, R, mixed-language, and deployment workflows







# Teachers' notes

Containers need to be able to talk to the host operating system and its kernel in order to execute their computations. A container isolates processes (OS level).

VM hypervisors usually isolate the host operating completely and allow the VM operating system to directly address the host hardware using its own kernel, this **mainly increases system-level reproducibility, not numerical precision** (good for file system application and legacy software, little impact on reproducibility)

‘Special case’: A container image may contain a complete OS user space, but standard containers do not include their own running kernel. They share the kernel of the host—or of the VM serving as the Kubernetes node.

# Intended Learning Outcomes

- Define what computational reproducibility is and how this is achieved
- Understand what virtual environments, containers, and virtual machines are and explain how they differ

# Intended Learning Outcomes

- Set a computational environment
- Containerised an application/analysis

Let's head to VS code and play with uv (env and dependencies together), the same exists for R (rig+renv) and Docker